

## Research Interests

My Ph.D. research bridged the gap between rough freehand line drawings and precise digital representations, as well as generating line drawings from 3D models. I am now expanding into generative image models and attribution, focusing on creative technologies that assist artists. My work spans 2D/3D geometry, vector graphics, non-photorealistic rendering, and analyzing LoRA models fine-tuned to mimic artistic styles. This aligns with my goal of integrating technical advances into artistic processes to foster a collaborative relationship between technology and creativity.

## Education

|                     |                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| Sep 2016 – Jun 2023 | <b>University of British Columbia</b><br>Ph.D. in Computer Graphics. Advised by Prof. Alla Sheffer. |
| Aug 2013 – Dec 2014 | <b>Carnegie Mellon University</b><br>M.S. in Computer Science.                                      |
| Sep 2009 – Jun 2013 | <b>Beihang University</b><br>B.Eng. in Computer Science and Technology. Ranking: 2/188.             |

## Publications

**Chenxi Liu**, Towaki Takikawa, and Alec Jacobson (2024). *A LoRA is Worth a Thousand Pictures*. Under review. arXiv: 2412.12048 [cs.CV]

**Chenxi Liu**, Siqi Wang, Matthew Fisher, Deepali Aneja, and Alec Jacobson (2024). *2D Neural Fields with Learned Discontinuities*. Eurographics'25. arXiv: 2408.00771 [cs.CV]

Siqi Wang, **Chenxi Liu**, Daniele Panozzo, Denis Zorin, and Alec Jacobson (2023). "Bézier Spline Simplification Using Locally Integrated Error Metrics". In: *SIGGRAPH Asia 2023 Conference Papers*. ACM

**Chenxi Liu** (2023). "Processing Freehand Vector Sketches". PhD thesis. University of British Columbia

**Chenxi Liu**, Toshiki Aoki, Mikhail Bessmeltsev, and Alla Sheffer (July 2023). "StripMaker: Perception-Driven Learned Vector Sketch Consolidation". In: *ACM Trans. Graph.* 42.4. SIGGRAPH'23

**Chenxi Liu**, Pierre Bénard, Aaron Hertzmann, and Shayan Hoshyari (Feb. 2023). "ConTesse: Accurate Occluding Contours for Subdivision Surfaces". In: *ACM Trans. Graph.* 42.1. SIGGRAPH'23

Jerry Yin, **Chenxi Liu**, Rebecca Lin, Nicholas Vining, Helge Rhodin, and Alla Sheffer (2022). "Detecting Viewer-Perceived Intended Vector Sketch Connectivity". In: *ACM Trans. Graph.* 41.4. SIGGRAPH'22

Dave Pagurek van Mossel, **Chenxi Liu**, Nicholas Vining, Mikhail Bessmeltsev, and Alla Sheffer (2021). "StrokeStrip: Joint Parameterization and Fitting of Stroke Clusters". In: *ACM Trans. Graph.* 40.4. SIGGRAPH'21

Yulia Gryaditskaya, Felix Hähnlein, **Chenxi Liu**, Alla Sheffer, and Adrien Bousseau (2020). “Lifting Freehand Concept Sketches into 3D”. in: *ACM Trans. Graph.* 39.6. SIGGRAPH Asia’20, pp. 1–16

**Chenxi Liu**, Enrique Rosales, and Alla Sheffer (2018). “StrokeAggregator: Consolidating Raw Sketches into Artist-Intended Curve Drawings”. In: *ACM Trans. Graph.* 37.4. SIGGRAPH’18

**Chenxi Liu**, Jessica Hodgins, and James McCann (July 2017). “Whole-cloth Quilting Patterns from Photographs”. In: *Proceedings of the Symposium on Non-Photorealistic Animation and Rendering*. NPAR’17

Lea Albaugh, April Grow, **Chenxi Liu**, James McCann, Gillian Smith, and Jennifer Mankoff (2016). “Threadsteading: Playful Interaction for Textile Fabrication Devices”. In: *Proceedings of the 2016 CHI Conference Extended Abstracts*. CHI’16. ACM

## Research & Professional Experience

|                     |                                                                                                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jul 2023 – Present  | <b>DGP Lab, University of Toronto</b><br>Postdoctoral Fellow. Supervised by Prof. Alec Jacobson.                                                                                                                                               |
| Mar 2023 – Jun 2023 | <b>Adobe Research</b><br>Research Intern. Supervised by Deepali Aneja and Prof. Alec Jacobson.                                                                                                                                                 |
| Sep 2016 – Feb 2023 | <b>Digital Geometry Processing Group, Imager Lab, UBC</b><br>Research Assistant. Advised by Prof. Alla Sheffer.<br>Conducting research on the artistic creation tools.                                                                         |
| May 2020 – Nov 2020 | <b>Adobe Research</b><br>Research Intern. Supervised by Aaron Hertzmann.<br>Conducted research on non-photorealistic line drawing generation.                                                                                                  |
| Mar 2015 – Jul 2016 | <b>Textile Lab, Disney Research Pittsburgh</b><br>Research Associate.<br>Supervised by Prof. James McCann & Prof. Jessica Hodgins.<br>Conducted research on automatic quilting pattern generation using CNC quilting machine and graph theory. |
| Jul 2014 – Aug 2014 | <b>CMU Graphics Lab</b><br>Research Assistant. Advised by Prof. Kayvon Fatahalian.<br>Built a visualization module for a lighting control framework using Arnold.                                                                              |
| Jul 2012 – Sep 2012 | <b>Microsoft Search Technology Center Asia</b><br>Intern, Software Development Engineer in Test.<br>Wrote scripts to gather statistical data and create analyses from search engine logs.                                                      |
| Aug 2011 – Dec 2011 | <b>Laboratory for Information Security and Intelligent Information Processing, Beihang University</b><br>Research Assistant. Advised by Prof. Zhoujun Li.<br>Built a tool to capture software exception messages.                              |

## Honors

|                      |                                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Apr 2024             | <b>Eurographics 2024 PhD Thesis Award, Honorable Mention</b><br>Eurographics and the Computer Graphics Forum Journal.    |
| Feb 2024             | <b>Faculty of Arts &amp; Science Postdoctoral Fellowship</b><br>Faculty of Arts & Science, University of Toronto.        |
| May 2022             | <b>WiGRAPH Rising Stars</b><br>The ACM Community Group for Women in Computer Graphics Research.                          |
| Oct 2016             | <b>Technology Award Winner: Threadsteading</b><br>IndieCade'16.                                                          |
| Nov 2011             | <b>National Scholarship (Top 1% in Academic Performance)</b><br>Ministry of Education of the People's Republic of China. |
| Dec 2012, 2011, 2010 | <b>The First Prize Scholarship of Academic Performance</b><br>Beihang University.                                        |

## Talks & Exhibition

|           |                                                                                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun 2024  | <b>SGP'24 Graduate School</b><br>Course: Fundamentals and Applications of Sketch Processing.                                                                                      |
| Aug 2023  | <b>SIGGRAPH'23</b><br>Conference Presenter: ConTesse: Accurate Occluding Contours for Subdivision Surfaces.<br>StripMaker: Perception-driven Learned Vector Sketch Consolidation. |
| Fall 2022 | <b>University of Toronto, Université de Montréal, University of Surrey</b><br>Invited Talk: Cleaning Up Vector Sketches Made by Humans and Computers.                             |
| Aug 2022  | <b>SIGGRAPH'22</b><br>Conference Presenter: Detecting Viewer-Perceived Intended Vector Sketch Connectivity.                                                                       |
| Aug 2018  | <b>SIGGRAPH'18</b><br>Conference Presenter: Strokeaggregator: Consolidating raw sketches into artist-intended curve drawings.                                                     |
| Jul 2017  | <b>NPAR'17</b><br>Conference Presenter: Whole-cloth quilting patterns from photographs.                                                                                           |
| Mar 2016  | <b>Alt.Ctrl.GDC, Game Developers Conference</b><br>Exhibitor: Threadsteading.<br>(Game also attended CHI Interactivity, 2016. IndieCade, 2016)                                    |

## Teaching Experience

|                |                                                                                                                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fall, 2023     | <b>CSC317@UofT: Computer Graphics</b><br>Guest Lecturer and Co-Developer of LEAF+ Project (Understanding the Limits of AI-Based Image Generators with DALL·E 2 and Midjourney). Project Lead: Prof. Alec Jacobson. |
| Winter 1, 2022 | <b>CPSC424@UBC: Geometric Modeling</b><br>Graduate Teaching Assistant, Instructor: Prof. Alla Sheffer.                                                                                                             |
| Winter, 2019   | <b>Instructional Skills Workshops for Grad Students@UBC</b>                                                                                                                                                        |
| Winter 2, 2018 | <b>CPSC436D@UBC: Video Game Programming</b><br>Graduate Teaching Assistant, Instructor: Prof. Alla Sheffer.                                                                                                        |
| Winter 2, 2016 | <b>CPSC418@UBC: Parallel Computation</b><br>Graduate Teaching Assistant, Instructor: Prof. Mark Greenstreet.                                                                                                       |

## Undergraduate Advising

|             |                                              |
|-------------|----------------------------------------------|
| 2023 – 2024 | Sepehr Ghasemipour<br>University of Toronto. |
| 2023 – 2024 | Silvia Lopez<br>University of Toronto.       |
| 2022 – 2023 | Toshiki Aoki<br>University of Tokyo.         |

## Services

|                |                                                                                                                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024           | <b>DGP Academy</b><br>Project Lead: Low-Poly Fabrication. Media.                                                                                                                                     |
| 2019 – Present | <b>Conference and Journal Paper Review</b><br>SIGGRAPH'22,23,24. SIGGRAPH Asia'22,24. Eurographics'20,22,23,24. Pacific Graphics'20,24. TVCG'23. CGF'22,24. IEEE CG&A'20. TPAMI'19. SCF'19. UIST'24. |
| 2018           | <b>Graduate Student Recruiting Committee</b><br>Student Representative of Computer Graphics.                                                                                                         |
| 2016 – 2017    | <b>AMORE Seminar</b><br>Imager Lab seminar organizer.                                                                                                                                                |

## Skills

Programming: C/C++, Python, MATLAB, Java, SQL.

Libraries and Tools: libigl, CGAL, scikit-learn, CUDA, OpenCL, OpenGL, CMake, Git.

Visual Editing: Illustrator, Photoshop, Premiere.

## References

Alec Jacobson (Postdoc supervisor, [jacobson@cs.toronto.edu](mailto:jacobson@cs.toronto.edu)), University of Toronto.

Alla Sheffer (Ph.D. supervisor, [sheffa@cs.ubc.ca](mailto:sheffa@cs.ubc.ca)), University of British Columbia.

Aaron Hertzmann (internship mentor, [hertzman@dgp.toronto.edu](mailto:hertzman@dgp.toronto.edu)), Adobe Research.